Zolbetuximab for Unresectable and Metastatic Gastric and Gastroesophageal Junction Adenocarcinoma: A Review of Literature.

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This article comprehensively reviews the working, efficacy, and safety profile of zolbetuximab. Zolbetuximab is a pioneering chimeric monoclonal antibody designed to target Claudin 18.2 (CLDN18.2), a tight junction protein frequently overexpressed in various gastrointestinal cancers, including gastric (G) and gastroesophageal junction (GEJ) adenocarcinomas. This drug has captured attention in the treatment of unresectable and metastatic G/GEJ cancer, especially in HER2-negative patients whose tumours express CLDN18.2. Zolbetuximab is a drug that binds to CLDN18.2, and its binding initiates an immune response that attacks and kills the cancer cells. It is typically co-prescribed with fluoropyrimidine and platinum-containing chemotherapy. The drug (formerly IMAB362), brand name Vyloy, was developed by Astellas Pharma, Tokyo, Japan. After multiple rounds of clinical trials, it was approved by the U.S. Food and Drug Administration (FDA) as a first-line treatment for locally advanced, unresectable cancer, establishing itself as a promising option for advanced G/GEJ cancers. Studies reveal that it targets CLDN18.2 for the selective killing of cancer cells while sparing most healthy tissues. Zolbetuximab has demonstrated a significant improvement in progression-free survival (PFS) when combined with chemotherapy (like mFOLFOX6 or CAPOX). Also, it showed prolonged PFS compared to chemotherapy alone in previously untreated, CLDN18.2-positive, HER2-negative patients. Zolbetuximab has also shown improvements in overall survival (OS), regardless of whether the cancer progresses. This is a crucial benefit for patients with advanced G/GEJ adenocarcinomas. Additionally, zolbetuximab's dual action triggers antibody-dependent cellular cytotoxicity (ADCC) and complement-dependent cytotoxicity (CDC). This enhances the immune system's ability to destroy cancerous cells and results in highly potent tumour destruction compared with chemotherapy alone. Zolbetuximab is a relatively safe drug with a few adverse effects. The most frequently reported side effects were gastrointestinal, namely nausea, fatigue, vomiting, decreased appetite, diarrhea, neutropenia, anemia, and thrombocytopenia, potentially due to specific CLDN18.2 expression in the gastric mucosa. Its side effects are generally manageable, with no unexpected toxicity beyond those typically seen in patients receiving chemotherapy.

Effect of anti-claudin 18.2 monoclonal antibody zolbetuximab alone or combined with chemotherapy or programmed cell death-1 blockade in syngeneic and xenograft gastric cancer models.

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The development of targeted cancer therapies based on monoclonal antibodies against tumor-associated antigens has progressed markedly over recent decades. This approach is dependent on the identification of tumor-specific, normal tissue-sparing antigenic targets. The transmembrane protein claudin-18 splice variant 2 (CLDN18.2) is frequently and preferentially displayed on the surface of primary gastric adenocarcinomas, making it a promising monoclonal antibody target. Phase 3 studies of zolbetuximab, a chimeric immunoglobulin G1 monoclonal antibody targeting CLDN18.2, combined with 5-fluorouracil/leucovorin plus oxaliplatin (modified FOLFOX6) or capecitabine plus oxaliplatin (CAPOX) in advanced or metastatic first-line gastric or gastroesophageal junction (G/GEJ) adenocarcinoma have demonstrated favorable clinical results with zolbetuximab. In studies using xenograft or syngeneic models with gastric cancer cell lines, zolbetuximab mediated death of CLDN18.2-positive human cancer cell lines via antibody-dependent cellular cytotoxicity and complement-dependent cytotoxicity in vitro and demonstrated anti-tumor efficacy as monotherapy and combined with chemotherapy in vivo. Mice treated with zolbetuximab plus chemotherapy displayed a significantly higher frequency of tumor-infiltrating CD8+ T cells versus vehicle/isotype control-treated mice. Furthermore, zolbetuximab combined with an anti-mouse programmed cell death-1 antibody more potently inhibited tumor growth compared with either agent alone. These results support the potential of zolbetuximab as a novel treatment option for G/GEJ adenocarcinoma.

[A New Molecular Targeted Agent for Gastric Cancer-The Anti-Claudin 18.2 Antibody, Zolbetuximab].

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The standard first-line treatment for unresectable advanced or recurrent gastric cancer(GC)and gastroesophageal junction cancer(GEJC)has been a platinum doublet chemotherapy. Trastuzumab with chemotherapy is the standard regimen for HER2-positive GC/GEJC. While, for HER2-negative cases, chemotherapy with or without immune checkpoint inhibitors (ICIs)such as nivolumab or pembrolizumab are regarded as the standard therapy. However, many patients do not derive benefit from anti-HER2 targeted therapies or ICIs, and new therapeutic targets have been explored. CLDN18.2 has emerged as a tissue-specific therapeutic target in gastric cancer. Zolbetuximab, a first-in-class chimeric IgG1 monoclonal antibody targeting CLDN18.2, has been developed. Zolbetuximab induces cancer cell death through antibody-dependent cellular cytotoxicity(ADCC)and complement-dependent cytotoxicity(CDC). Recently, zolbetuximab with chemotherapy improved survival rates in HER2-negative CLDN18.2-positive previously untreated patients with unresectable advanced or recurrent GC/GEJC, leading to its approval in Japan and its establishment as a standard treatment. GC/GEJC with early onset, scirrhous type or peritoneum dissemination commonly express CLDN18.2. The emergence of this novel therapeutic option is of great significance in clinical practice. We will highlight the previous clinical trials of zolbetuximab and provide future perspective of CLDN18.2 targeted therapy. In addition, we will introduce the ongoing development of various CLDN18.2 targeting therapies such as newer monoclonal antibodies, antibody-drug conjugates(ADCs), chimeric antigen receptor T-cell(CAR- T)therapy, and bispecific antibodies.

Characterization of zolbetuximab in pancreatic cancer models.

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In healthy tissue, the tight junction protein Claudin 18.2 (CLDN18.2) is present only in the gastric mucosa. Upon malignant transformation of gastric epithelial tissue, perturbations in cell polarity lead to cell surface exposure of CLDN18.2 epitopes. Moreover, CLDN18.2 is aberrantly expressed in malignancies of several other organs, such as pancreatic cancer (PC). A monoclonal antibody, zolbetuximab (formerly known as IMAB362), has been generated against CLDN18.2. In a phase 2 clinical trial (FAST: NCT01630083), zolbetuximab in conjunction with chemotherapy prolonged overall and progression-free survival over chemotherapy alone and improved quality of life. In this study, the mechanism of action and antitumor activity of zolbetuximab were investigated using nonclinical PC models. Zolbetuximab bound specifically and with strong affinity to human PC cells that expressed CLDN18.2 on the cell surface. In ex vivo systems using immune effector cells and serum from healthy donors, zolbetuximab induced antibody-dependent cellular cytotoxicity (ADCC) and complement-dependent cytotoxicity (CDC), resulting in the lysis of cultured human PC cells. The amplitude of ADCC and CDC directly correlated with cell surface CLDN18.2 levels. The chemotherapeutic agent gemcitabine upregulated CLDN18.2 expression in cultured human PC cells and enhanced zolbetuximab-induced ADCC. In mouse xenograft tumors derived from human PC cell lines, including gemcitabine-refractory ones, zolbetuximab slowed tumor growth, benefited survival, and attenuated metastases development. The results presented here validate CLDN18.2 as a targetable biomarker in PC and support extension of the clinical development of zolbetuximab to patients with CLDN18.2-expressing PC.

A phase I dose-escalation study of IMAB362 (Zolbetuximab) in patients with advanced gastric and gastro-oesophageal junction cancer.

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IMAB362 (Zolbetuximab) is a chimeric monoclonal antibody that binds to Claudin-18.2, a target antigen specific to cancer cells. In vitro, IMAB362 mediates cell death through antibody-dependent cellular cytotoxicity and complement-dependent cytotoxicity; thus, IMAB362 may serve as a potent, targeted immunotherapeutic agent. This first-in-human phase I study enroled adult patients (N = 15) with advanced gastric or gastro-oesophageal junction cancer into five sequential single dose-escalation cohorts (33, 100, 300, 600, and 1000 mg/m2) following a 3 + 3 design. Safety/tolerability, including determination of maximum tolerated dose and recommended phase II dose, were the primary objectives; secondary objectives included assessment of the IMAB362 pharmacokinetic profile, immunogenicity, and antitumour activity (assessed by Response Evaluation Criteria in Solid Tumors v1.0). IMAB362 was generally well tolerated at all doses, with gastrointestinal toxicities being the most commonly observed treatment-related adverse events. As dose-limiting toxicity was not observed within 4 weeks of treatment, a maximum tolerated dose was not established. The pharmacokinetic profile of IMAB362 appeared to be proportional across the dose range; and mean half-life ranged from 13 to 24 d. While most patients showed progressive disease at weeks 4-5 after a single intravenous IMAB362 infusion, one patient in the 600 mg/m2 dose group achieved and maintained stable disease for approximately 2 months postinfusion. Findings from this study demonstrate that IMAB362 is generally well tolerated and support further evaluation in patients with gastric/gastro-oesophageal junction cancer. ClinicalTrials.gov, Identifier NCT00909025.

Phase 1 trial of zolbetuximab in Japanese patients with CLDN18.2+ gastric or gastroesophageal junction adenocarcinoma.

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Zolbetuximab is a chimeric monoclonal antibody that targets claudin-18.2, a candidate biomarker in patients with advanced gastric/gastroesophageal cancer. This nonrandomized phase 1 study (NCT03528629) enrolled previously treated Japanese patients with claudin-18.2-positive locally advanced/metastatic gastric/gastroesophageal cancer in two parts: Safety (Arms A and B, n = 3 each) and Expansion (n = 12). Patients received intravenous zolbetuximab 800 mg/m2 on cycle 1, day 1 followed by 600 mg/m2 every 3 weeks (Q3W; Safety Part Arm A and Expansion) or 1000 mg/m2 Q3W (Safety Part Arm B). For the Safety Part, the primary endpoint was safety (i.e., dose-limiting toxicities

[DLTs]) and a secondary endpoint was objective response rate (ORR) by investigator. For the Expansion Part, the primary endpoint was ORR by investigator and secondary endpoints included ORR by central review and safety. Additional secondary endpoints for both the Safety and Expansion Parts were disease control rate (DCR), overall survival (OS), progression-free survival (PFS), duration of response, pharmacokinetics, and immunogenicity. In 18 patients, no DLTs (Safety Part) or drug-related treatment-emergent adverse events (TEAEs) grade ≥3 were observed. Most TEAEs were gastrointestinal. In 17 patients with measurable lesions, best overall response was stable disease (64.7%) or progressive disease (35.3%). The DCR was 64.7% (95% confidence interval 38.3-85.8). In Arm A and Expansion combined (n = 15), median OS was 4.4 months (2.6-11.4) and median PFS was 2.6 months (0.9-2.8). In Arm B (n = 3), median OS was 6.4 months (2.9-6.8) and median PFS was 1.7 months (1.2-2.1). Zolbetuximab exhibited no new safety signals with limited single-agent activity in Japanese patients.

Claudin 18.2 is a potential therapeutic target for zolbetuximab in pancreatic ductal adenocarcinoma.

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Pancreatic ductal adenocarcinoma (PDAC) is frequently diagnosed and treated in advanced tumor stages with poor prognosis. More effective screening programs and novel therapeutic means are urgently needed. Recent studies have regarded tight junction protein claudin 18.2 (CLDN18.2) as a candidate target for cancer treatment, and zolbetuximab (formerly known as IMAB362) has been developed against CLDN18.2. However, there are few data reported thus far related to the clinicopathological characteristics of CLDN18.2 expression for PDAC. To investigate the expression of CLDN18.2 in PDAC patients and subsequently propose a new target for the treatment of PDAC. The Cancer Genome Atlas, Genotype-Tissue Expression, Gene Expression Omnibus, and European Genome-phenome Archive databases were first employed to analyze the CLDN18 gene expression in normal pancreatic tissue compared to that in pancreatic cancer tissue. Second, we analyzed the expression of CLDN18.2 in 93 primary PDACs, 86 para-cancer tissues, and 13 normal pancreatic tissues by immunohistochemistry. Immunostained tissues were assessed applying the histoscore. subsequently, they fell into two groups according to the expression state of CLDN18.2. Furthermore, the correlations between CLDN18.2 expression and diverse clinicopathological characteristics, including survival, were investigated. The gene expression of CLDN18 was statistically higher ($P < 0.01$) in pancreatic tumors than in normal tissues. However, there was no significant correlation between CLDN18 expression and survival in pancreatic cancer patients. CLDN18.2 was expressed in 88 (94.6%) of the reported PDACs. Among these tumors, 50 (56.8%) cases showed strong immunostaining. The para-cancer tissues were positive in 81 (94.2%) cases, among which 32 (39.5%) of cases were characterized for strong staining intensities. Normal pancreatic tissue was identified solely via weak immunostaining. Finally, CLDN18.2 expression significantly correlated with lymph node metastasis, distant metastasis, nerve invasion, stage, and survival of PDAC patients, while there was no correlation between CLDN18.2 expression and localization, tumor size, patient age and sex, nor any other clinicopathological characteristic. CLDN18.2 expression is frequently increased in PDAC patients. Thus, it may act as a potential therapeutic target for zolbetuximab in PDAC.

Zolbetuximab plus CAPOX in CLDN18.2-positive gastric or gastroesophageal junction

adenocarcinoma: the randomized, phase 3 GLOW trial.

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There is an urgent need for first-line treatment options for patients with human epidermal growth factor receptor 2 (HER2)-negative, locally advanced unresectable or metastatic gastric or gastroesophageal junction (mG/GEJ) adenocarcinoma. Claudin-18 isoform 2 (CLDN18.2) is expressed in normal gastric cells and maintained in malignant G/GEJ adenocarcinoma cells. GLOW (closed enrollment), a global, double-blind, phase 3 study, examined zolbetuximab, a monoclonal antibody that targets CLDN18.2, plus capecitabine and oxaliplatin (CAPOX) as first-line treatment for CLDN18.2-positive, HER2-negative, locally advanced unresectable or mG/GEJ adenocarcinoma. Patients (n = 507) were randomized 1:1 (block sizes of two) to zolbetuximab plus CAPOX or placebo plus CAPOX. GLOW met the primary endpoint of progression-free survival (median, 8.21 months versus 6.80 months with zolbetuximab versus placebo; hazard ratio (HR) = 0.687; 95% confidence interval (CI), 0.544-0.866; P = 0.0007) and key secondary endpoint of overall survival (median, 14.39 months versus 12.16 months; HR = 0.771; 95% CI, 0.615-0.965; P = 0.0118). Grade ≥3 treatment-emergent adverse events were similar with zolbetuximab (72.8%) and placebo (69.9%). Zolbetuximab plus CAPOX represents a potential new first-line therapy for patients with CLDN18.2-positive, HER2-negative, locally advanced unresectable or mG/GEJ adenocarcinoma. ClinicalTrials.gov identifier: NCT03653507 .

Zolbetuximab: First Approval.

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Zolbetuximab (VYLOY™), a recombinant, chimeric, anti-claudin 18.2 (CLDN18.2) monoclonal antibody (mAb), is being developed by Astellas Pharma Inc. for the treatment of patients with HER2-negative (HER2-), CLDN18.2-positive (CLDN18.2+) advanced gastric or gastroesophageal junction (GEJ) adenocarcinoma and CLDN18.2+ advanced pancreatic adenocarcinoma. In March 2024, zolbetuximab was approved in Japan for the treatment of patients with HER2-, CLDN18.2+ unresectable, advanced/recurrent gastric cancer (the gastric cancer indication includes GEJ cancer). Zolbetuximab is also undergoing regulatory review for HER2-, CLDN18.2+ advanced gastric or GEJ adenocarcinoma in the USA, the EU, China, Australia and several other countries. This article summarizes the milestones in the development of zolbetuximab leading to this first approval for the treatment of patients with CLDN18.2+ gastrointestinal malignancies.

Targeted and combination immunotherapies using biologics for gastric cancer: the state-of-the-art.

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Gastric adenocarcinoma (GAC) remains a prevalent cancer worldwide and its incidence is increasing in South America. The heterogenous nature of GAC makes advances in management challenging. Despite challenges, recent therapeutic targets are individualizing treatment. For localized disease with microsatellite-instability-high/deficient mismatch repair, immunotherapy is now an adopted practice. In the advanced unresectable setting, those harboring human epidermal growth factor receptor-2 (HER2) expression continue to be a separate entity. Future targets are developing. Among these include claudin

18.2 (CLDN18.2), fibroblast growth factor receptor 2b (FGFR2b), and trophoblast cell surface antigen-2 (TROP-2). FDA approval of zolbetuximab's, an anti-CLDN 18.2 monoclonal antibody, is expected soon. Additionally, bemarituzumab, ananti-FGFR2b monoclonal antibody, has shown improvements in combination with chemotherapy in those with HER2 negative GAC with FGFR2 overexpression. This combination is now being investigated in a phase 3 trial. Lastly, TROP-2 has emerged as an exciting solid tumor target and study is expected in GAC. All three of these therapeutic targets have seen an abundance of drug development in recent years, and we anticipate newer targeted agents driving therapeutic decisions in GAC in the coming years.